

ASSIGNMENT - 3

i) Determine the multiplicative inverse modulo 26 for each of the following integers.

1, 3, 5, 7, 9, 11, 15, 17, 19, 21, 23, 25

i) 3^{-1}

$$26 = 3(8) + 2$$

$$3 = 2(1) + 1$$

$$1 = 3 + 2(-1)$$

$$1 = 3 + [26 + 3(-8)](-1)$$

$$1 = 3 + 26(-1) + 3(8)$$

$$3(9) + 26(-1) = 1$$

$$\boxed{3^{-1} = 9}$$

$$11 = (1-1)21 + 25$$

$$11 = (1-1)11 + 21$$

$$2 = (5-1)11 + 11$$

$$1 = (1-1)5 + 11$$

$$11 = (1-1)21 + 25$$

$$11 = (1-1)11 + 21$$

$$2 = (5-1)11 + 11$$

$$1 = (1-1)5 + 11$$

ii) 5^{-1}

$$26 = 5(5) + 1$$

$$26 + 5(-5) = 1$$

$$26 - 5 = 21$$

$$\boxed{5^{-1} = 21}$$

$$11 = (1-1)[(5-1)11 + 11] + (1-1)11 + 21$$

$$11 = (5-1)11 + (1-1)11 + (1-1)11 + 21$$

$$11 = (5-1)11 + (5-1)11 + 21$$

$$1 = (5)[(1-1)11 + 21] + (5-1)[(1-1)21 + 25] + 11$$

$$1 = (5-1)11 + (5)21 + (5-1)21 + (5-1)25 + 21$$

$$1 = (5-1)[(1-1)21 + 25] + (5)21 + (5-1)21$$

$$1 = (5)21 + (5-1)21 + (5-1)25 + (5-1)21$$

$$1 = (11-1)21 + (1-1)21$$

$$\boxed{7^{-1} = 15}$$

iii) 7^{-1}

$$26 = 7(3) + 5$$

$$7 = 5(1) + 2$$

$$5 = 2(2) + 1$$

$$26 + 7(-3) + [7 + 5(-1)](-2) = 1$$

$$26 + 7(-3) + 7(-2) + 5(2) = 1$$

$$26 + 7(-5) + 5(2) = 1$$

$$26 + 7(-5) + [26 + 7(-3)](2) = 1$$

$$26 + 7(-5) + 26(2) + 7(-6) = 1$$

$$26(3) + 7(-11) = 1$$

$$26 - 11 = 15$$

iv) 9^{-1}

$$26 = 9(2) + 8$$

$$9 = 8(1) + 1$$

$$8 = 26 + 9(-2)$$

$$9 + 8(-1) = 1$$

$$9 + [26 + 9(-2)](-1) = 1$$

$$9 + [26(-1) + 9(2)] = 1$$

$$9(3) + 26(-1) = 1$$

$$\boxed{9^{-1} = 3}$$

v) 11^{-1}

$$26 = 11(2) + 4$$

$$4 = 26 + 11(-2)$$

$$11 = 4(2) + 3$$

$$3 = 11 + 4(-2)$$

$$4 = 3(1) + 1$$

$$1 = 4 + 3(-1)$$

$$26 + 11(-2) + [11 + 4(-2)](-1) = 1$$

$$26 + 11(-2) + 11(-1) + 4(2) = 1$$

$$26 + 11(-3) + [26 + 11(-2)](2) = 1$$

$$26 + 11(-3) + 26(2) + 11(-4) = 1$$

$$26(3) + 11(-7) = 1$$

$$26 - 7 = 19$$

$$\boxed{11^{-1} = 19}$$

vi) 15^{-1}

$$26 = 15(1) + 11$$

$$26 + 15(-1) = 11$$

$$15 = 11(1) + 4$$

$$15 + 11(-1) = 4$$

$$11 = 4(2) + 3$$

$$11 + 4(-2) = 3$$

$$4 = 3(1) + 1$$

$$4 + 3(-1) = 1$$

$$15 + 11(-1) + [11 + 4(-2)](-1) = 1$$

$$15 + 11(-1) + 11(-1) + 4(2) = 1$$

$$15 + 11(-2) + 4(2) = 1$$

$$15 + [26 + 15(-1)](-2) + [15 + 11(-1)](2) = 1$$

$$15 + 26(-2) + 15(2) + 15(2) + 11(-2) = 1$$

$$15(5) + 26(-2) + [26 + 15(-1)](-2) = 1$$

$$15(5) + 26(-2) + 26(-2) + 15(2) = 1$$

$$15(7) + 26(-4) = 1$$

$$\boxed{15^{-1} = 7}$$

vii) 17^{-1}

$$26 = 17(1) + 9$$

$$9 = 26 + 17(-1)$$

$$17 = 9(1) + 8$$

$$8 = 17 + 9(-1)$$

$$9 = 8(1) + 1$$

$$1 = 9 + 8(-1)$$

$$26 + 17(-1) + [17 + 9(-1)](-1) = 1$$

$$26 + 17(-1) + 17(-1) + 9(1) = 1$$

$$26 + 17(-1) + 17(-1) + 26 + 17(-1) = 1$$

$$26(2) + 17(-3) = 1$$

$$26 - 3 = 23$$

$$\boxed{17^{-1} = 23}$$

viii) 19^{-1}

$$26 = 19(1) + 7$$

$$19 = 7(2) + 5$$

$$7 = 5(1) + 2$$

$$5 = 2(2) + 1$$

$$7 = 26 + 19(-1)$$

$$5 = 19 + 7(-2)$$

$$2 = 7 + 5(-1)$$

$$1 = 5 + 2(-2)$$

$$19 + 7(-2) + [7 + 5(-1)](-2) = 1$$

$$19 + [26 + 19(-1)](-2) + [26 + 19(-1)](-2) + 5(2) = 1$$

$$19 + 26(-2) + 19(2) + 26(-2) + 19(2) + [19 + 7(-2)](2) = 1$$

$$19(5) + 26(-4) + 19(2) + 7(-4) = 1$$

$$19(7) + 26(-4) + [26 + 19(-1)](-4) = 1$$

$$19(7) + 26(-4) + 26(-4) + 19(4) = 1$$

$$19(11) + 26(-8) = 1$$

$$\boxed{19^{-1} = 11}$$

ix) 21^{-1}

$$26 = 21(1) + 5$$

$$21 = 5(4) + 1$$

$$5 = 26 + 21(-1)$$

$$1 = 21 + 5(-4)$$

$$21 + [26 + 21(-1)](-4) = 1$$

$$21 + 26(-4) + 21(4) = 1$$

$$21(5) + 26(-4) = 1$$

$$\boxed{21^{-1} = 5}$$

x) 23^{-1}

$$26 = 23(1) + 3$$

$$23 = 3(7) + 2$$

$$3 = 2(1) + 1$$

$$3 = 26 + 23(-1)$$

$$2 = 23 + 3(-7)$$

$$26 + 23(-1) + [23 + 3(-7)](-1) = 1$$

$$26 + 23(-1) + 23(-1) + 3(7) = 1$$

$$26 + 23(-2) + 3(7) = 1$$

$$26 + 23(-2) + [26 + 23(-1)](7) = 1$$

$$26 + 23(-2) + 26(7) + 23(-7) = 1$$

$$26(8) + 23(-9) = 1$$

$$26 - 9 = 17$$

$$\boxed{23^{-1} = 17}$$

xi) 25^{-1}

$$26 = 25(1) + 1$$

$$26 + 25(-1) = 1$$

$$26 - 1 = 25$$

$$\boxed{25^{-1} = 25}$$

2) Choose any string of your choice to encrypt using the affine cipher with key values : multiplier 5, & additive 8

$$C = (P * K_1 + K_2) \bmod 26$$

$$K_1 = 5, K_2 = 8$$

PT = GOWTHAM

$$G = (6 * 5 + 8) \bmod 26 \Rightarrow 38 \bmod 26 \Rightarrow 12 \Rightarrow M$$

$$O = (14 * 5 + 8) \bmod 26 \Rightarrow 78 \bmod 26 \Rightarrow 0 \Rightarrow A$$

$$W = (22 * 5 + 8) \bmod 26 \Rightarrow 118 \bmod 26 \Rightarrow 14 \Rightarrow O$$

$$T = (19 * 5 + 8) \bmod 26 \Rightarrow 103 \bmod 26 \Rightarrow 25 \Rightarrow Z$$

$$H = (7 * 5 + 8) \bmod 26 \Rightarrow 43 \bmod 26 \Rightarrow 17 \Rightarrow R$$

$$A = (0 * 5 + 8) \bmod 26 \Rightarrow 8 \bmod 26 \Rightarrow 8 \Rightarrow I$$

$$M = (12 * 5 + 8) \bmod 26 \Rightarrow 68 \bmod 26 \Rightarrow 16 \Rightarrow Q$$

CT = MAOZR IQ

3) Encrypt "HI" using Affine Cipher with $a=11$ and $b=5$

$$C = (P * a + b) \bmod 26$$

$$H = (7 * 11 + 5) \bmod 26 \Rightarrow 82 \bmod 26 \Rightarrow 4 \Rightarrow E$$

$$I = (8 * 11 + 5) \bmod 26 \Rightarrow 93 \bmod 26 \Rightarrow 15 \Rightarrow P$$

CT = EP

4) The cipher text "UCR" was encrypted using the affine function $9x + 2 \bmod 26$. Find the plaintext.

$$C = (9P + 2) \bmod 26$$

$$P = a^{-1} (C - b) \bmod 26$$

$$a = 9$$

$$b = 2$$

$$9^{-1} = 3$$

$$U = 3(20 - 2) \bmod 26 \Rightarrow 54 \bmod 26 \Rightarrow 2 \Rightarrow C$$

$$C = 3(2 - 2) \bmod 26 \Rightarrow 0 \bmod 26 \Rightarrow 0 \Rightarrow A$$

$$R = 3(17 - 2) \bmod 26 \Rightarrow 45 \bmod 26 \Rightarrow 19 \Rightarrow T$$

PT = CAT

5) Decrypt the cipher text using Affine cipher, 'KDYXTJWDEXZY' using the key 3 (Multiplier) and 4.

$$P = a^{-1}(C - b) \bmod 26$$

$$a = 3, b = 4$$

$$3^{-1} = 9$$

$$26 = 3(8) + 2$$

$$3 = 2(1) + 1$$

$$3 + [26 + 3(-8)](-1) = 1$$

$$3 + 26(-1) + 3(8) = 1$$

$$26(-1) + 3(9) = 1$$

$$3^{-1} = 9$$

$$K = 9(10 - 4) \bmod 26 \Rightarrow 54 \bmod 26 \Rightarrow 2 \Rightarrow C$$

$$D = 9(3 - 4) \bmod 26 \Rightarrow -9 \bmod 26 \Rightarrow (26 - 9) \Rightarrow 17 \Rightarrow R$$

$$Y = 9(24 - 4) \bmod 26 \Rightarrow 180 \bmod 26 \Rightarrow 24 \Rightarrow Y$$

$$X = 9(23 - 4) \bmod 26 \Rightarrow 171 \bmod 26 \Rightarrow 15 \Rightarrow P$$

$$J = 9(9 - 4) \bmod 26 \Rightarrow 45 \bmod 26 \Rightarrow 19 \Rightarrow T$$

$$V = 9(20 - 4) \bmod 26 \Rightarrow 144 \bmod 26 \Rightarrow 14 \Rightarrow O$$

$$W = 9(22 - 4) \bmod 26 \Rightarrow 162 \bmod 26 \Rightarrow 6 \Rightarrow L$$

$$D = 9(3 - 4) \bmod 26 \Rightarrow -9 \bmod 26 \Rightarrow (26 - 9) \Rightarrow 17 \Rightarrow R$$

$$E = 9(4 - 4) \bmod 26 \Rightarrow 0 \bmod 26 \Rightarrow 0 \Rightarrow G$$

$$X = 9(23 - 4) \bmod 26 \Rightarrow 171 \bmod 26 \Rightarrow 15 \Rightarrow P$$

$$Z = 9(25 - 4) \bmod 26 \Rightarrow 189 \bmod 26 \Rightarrow 7 \Rightarrow H$$

$$Y = 9(24 - 4) \bmod 26 \Rightarrow 180 \bmod 26 \Rightarrow 24 \Rightarrow Y$$

PT $\Rightarrow$ CRYPTOGRAPHY

6) Use $K_1 = 13$ (multiplier), $K_2 = 4$ (Even though it is not a valid key). Encrypt the string "crypt" and "alter" using the above key and find what is the issue.

$$C = (P \cdot K_1 + K_2) \bmod 26$$

$$K_1 = 13, K_2 = 4$$

13 is not coprime of 26

so we cannot find inverse of 13 mod 26. $\therefore$ it is not a valid key for affine cipher.

$$I \Rightarrow (8 \times 13 + 4) \bmod 26 \Rightarrow 108 \bmod 26 \Rightarrow 4 \Rightarrow E$$

$$N \Rightarrow (13 \times 13 + 4) \bmod 26 \Rightarrow 173 \bmod 26 \Rightarrow 17 \Rightarrow R$$

$$P \Rightarrow (15 \times 13 + 4) \bmod 26 \Rightarrow 199 \bmod 26 \Rightarrow 17 \Rightarrow R$$

$$U \Rightarrow (20 \times 13 + 4) \bmod 26 \Rightarrow 264 \bmod 26 \Rightarrow 4 \Rightarrow E$$

$$T \Rightarrow (19 \times 13 + 4) \bmod 26 \Rightarrow 251 \bmod 26 \Rightarrow 17 \Rightarrow R$$

$$\begin{aligned}
 A &\Rightarrow (0 \times 13 + 4) \bmod 26 \Rightarrow 4 \bmod 26 \Rightarrow 4 \Rightarrow E \\
 L &\Rightarrow (11 \times 13 + 4) \bmod 26 \Rightarrow 147 \bmod 26 \Rightarrow 17 \Rightarrow R \\
 T &\Rightarrow (19 \times 13 + 4) \bmod 26 \Rightarrow 251 \bmod 26 \Rightarrow 17 \Rightarrow R \\
 E &\Rightarrow (4 \times 13 + 4) \bmod 26 \Rightarrow 56 \bmod 26 \Rightarrow 4 \Rightarrow E \\
 R &\Rightarrow (17 \times 13 + 4) \bmod 26 \Rightarrow 225 \bmod 26 \Rightarrow 17 \Rightarrow R
 \end{aligned}$$

ERROR

Both the plain text will produce the same ciphertext. The encryption is not one to one and decryption is not possible as 13 is not a valid key.

7) Find the Inverse matrix for the key $\begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} \bmod 26$

$$K = \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} \bmod 26$$

: determinant of key

$$\begin{aligned}
 &= |3 \times 7 - 2 \times 5| \\
 &= |21 - 10| \\
 &= 11
 \end{aligned}$$

$$\begin{aligned}
 11^{-1} &\Rightarrow 26 = 11(2) + 4 & 4 &= 26 + 11(-2) \\
 &11 = 4(2) + 3 & 3 &= 11 + 4(-2) \\
 &4 = 3(1) + 1 & 1 &= 4 + 3(-1)
 \end{aligned}$$

$$[26 + 11(-2)] + [11 + 4(-2)](-1) = 1$$

$$26 + 11(-2) + 11(-1) + 4(2) = 1$$

$$26 + 11(-3) + 4(2) = 1$$

$$26 + 11(-3) + [26 + 11(-2)](2) = 1$$

$$26 + 11(-3) + 26(2) + 11(-4) = 1$$

$$26(3) + 11(-7) = 1$$

$$26 - 7 = 19$$

$$\boxed{11^{-1} = 19}$$

$$\text{Adjacent of matrix}(K) = \begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix}$$

$$K^{-1} = 19 \begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix} \bmod 26$$

$$K^{-1} = \begin{bmatrix} 133 & -38 \\ -95 & 57 \end{bmatrix} \bmod 26$$

$$K^{-1} = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix}$$

8) Find the Inverse of the following matrix whose entries are considered modulo 26. $\begin{bmatrix} 11 & 13 \\ 2 & 3 \end{bmatrix}$

$$\therefore K = \begin{bmatrix} 11 & 13 \\ 2 & 3 \end{bmatrix} \text{ mod } 26$$

$$D(K) = |11 \times 3 - 13 \times 2| = |33 - 26| = 7$$

$$7^{-1} \Rightarrow \begin{aligned} 26 &= 7(3) + 5 \\ 7 &= 5(1) + 2 \\ 5 &= 2(2) + 1 \end{aligned}$$

$$\Rightarrow 5 = 26 + 7(-3)$$

$$2 = 7 + 5(-1)$$

$$1 = 5 + 2(-2)$$

$$5 + [7 + 5(-1)](-2) = 1$$

$$26 + 7(-3) + 7(-2) + 5(2) = 1$$

$$26 + 7(-5) + [26 + 7(-3)](2) = 1$$

$$26 + 7(-5) + 26(2) + 7(-6) = 1$$

$$26(3) + 7(-11) = 1$$

$$26 - 11 = 15$$

$$7^{-1} = 15$$

$$\text{Adj}(K) = \begin{bmatrix} 3 & -13 \\ -2 & 11 \end{bmatrix}$$

$$K^{-1} = 15 \begin{bmatrix} 3 & -13 \\ -2 & 11 \end{bmatrix} \text{ mod } 26$$

$$K^{-1} = \begin{bmatrix} 45 & -195 \\ -30 & 165 \end{bmatrix} \text{ mod } 26$$

$$K^{-1} = \begin{bmatrix} 19 & 13 \\ 22 & 9 \end{bmatrix}$$

9) Is the key valid? $\begin{bmatrix} 2 & 14 \\ 13 & 21 \end{bmatrix}$

$$\therefore D(K) = |2 \times 21 - 13 \times 14| = |42 - 182| = -140$$

$$-140 \text{ mod } 26 = 16$$

$$\det K = 16 \text{ mod } 26$$

For a key to be valid its determinant value should be coprime with 26

$\therefore$ It is not a valid key

10) Encrypt using 2×2 Hill cipher for the plaintext
 "SOLVED" using $\begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix}$

$$\therefore K = \begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} \Rightarrow D(K) = |3 \times 11 - 8 \times 7| = |33 - 56| = -23 + 26 = 3$$

3 is coprime w

$$SOLVED \Rightarrow C = K \times P$$

$$SO \Rightarrow P = \begin{bmatrix} 18 \\ 14 \end{bmatrix} \begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} = \begin{bmatrix} 54 + 112 \\ 98 + 154 \end{bmatrix} = \begin{bmatrix} 166 \\ 280 \end{bmatrix} \pmod{26}$$

$$166 \pmod{26} \Rightarrow 10 \Rightarrow K$$

$$280 \pmod{26} \Rightarrow 20 \Rightarrow U$$

$$LV \Rightarrow P = \begin{bmatrix} 11 \\ 21 \end{bmatrix} \begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} = \begin{bmatrix} 33 + 168 \\ 77 + 231 \end{bmatrix} = \begin{bmatrix} 201 \\ 308 \end{bmatrix} \pmod{26}$$

$$201 \pmod{26} \Rightarrow 19 \Rightarrow T$$

$$308 \pmod{26} \Rightarrow 22 \Rightarrow W$$

$$ED \Rightarrow P = \begin{bmatrix} 4 \\ 3 \end{bmatrix} \begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} = \begin{bmatrix} 12 + 24 \\ 28 + 33 \end{bmatrix} = \begin{bmatrix} 36 \\ 61 \end{bmatrix} \pmod{26}$$

$$36 \pmod{26} \Rightarrow 10 \Rightarrow K$$

$$61 \pmod{26} \Rightarrow 9 \Rightarrow J$$

$$SOLVED \Rightarrow KUTWKJ$$

$$\boxed{\text{Cipher Text} = KUTWKJ}$$